Analysis of Classical and Quantum Phase Space Dynamics of the Kicked Top

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Abstract

In this work, we study the classical and quantum dynamics of the kicked top and reproduce the main theoretical results presented by Chaudhury, S. *et al.* [1]. The kicked top provides a useful model for studying the connection between classical chaos and quantum dynamics, for the parameters considered in this work, the classical kicked top exhibits a mixed phase space containing regular islands surrounded by a chaotic sea, providing a useful setting for studying the connection between classical chaos and quantum dynamics. The aim of this work is to theoretically model the kicked-top Hamiltonian corresponding to the experimental parameters used in the reference work and to reproduce the observed quantum phase-space dynamics.

Starting from the experimentally implemented Hamiltonian, the corresponding Floquet operator is derived and the classical map is obtained in the semiclassical limit. The resulting classical phase space is used to identify three different initial conditions: one inside the large regular island, one in the pair of smaller regular islands, and one in the chaotic sea. The corresponding spin coherent states are then evolved using Schrödinger evolution, and their dynamics are represented using the Husimi Q-function. The Floquet eigenstates and eigenphases are also calculated to understand the observed dynamics in terms of the eigenstate decomposition of the initial states.

The results show qualitatively different behaviour for the three initial conditions. The state in the large regular island remains localized, while the state in the small regular-island region shows dynamical tunnelling between the two islands. The tunnelling period is estimated from the difference between the relevant Floquet eigenphases and is found to be consistent with the Husimi evolution. The state initially placed in the chaotic sea explores a much larger region of phase space and exhibits a more rapidly varying fidelity. These results demonstrate the correspondence between the classical phase-space structure and the quantum dynamics of the kicked top, while also showing the quantum dynamical tunnelling that has no classical counterpart.

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Chapter 1

Introduction

Dynamics has been central to understanding complex systems in all fields of science, ranging from mechanics, population ecology, molecular dynamics etc to financial markets. While these systems are deterministic in nature and can be described by differential equations, and some of their behaviors can be predicted exactly, many of the systems in nature has non-linear interaction which give rise to remarkably intricate behavior which cannot be predicted over large timescales. This phenomenon is known as Deterministic chaos. Especially in classical physics, chaos theory has fundamentally reshaped our understanding by demonstrating simple equations of motion can give rise to irregular and seemingly random motion. Naturally this also raised fundamental questions regarding the presence of chaotic behavior in quantum regime, which itself is non-deterministic and doesn't have the notion of trajectories.

1.1 Classical Chaos

Classical mechanics describes the evolution of a dynamical system through equations of motion that determine its state at later times from its initial conditions. For linear and integrable systems, this evolution can often be understood and predicted for long times. However, nonlinear systems can depict a behaviour known as deterministic chaos. There is no single universally accepted definition of chaos, but a useful working definition given by Steven H Strogatz is that chaos is *aperiodic long-term behaviour in a deterministic system that exhibits sensitive dependence on initial conditions* [2].

The first important feature of chaos is its aperiodic long-term behaviour. A chaotic trajectory does not settle into a fixed point, a periodic orbit, or a quasiperiodic orbit as time increases. Instead, it continues to evolve in an

irregular manner. The second feature is determinism. Although the motion may appear random, there is no random force or noise required to produce it. The irregular behaviour arises from the nonlinear dynamics of the system. The third and most important feature is the sensitive dependence on initial conditions. Two trajectories that start arbitrarily close to each other can separate rapidly as the system evolves. This sensitivity is usually characterized by a positive Lyapunov exponent, which measures the average exponential rate at which nearby trajectories diverge [2].

A useful way to visualize these different types of dynamics is through phase space. A point in phase space represents the state of the system, and its evolution generates a trajectory. In a regular region, neighbouring trajectories remain confined to well defined structures, such as closed periodic or quasiperiodic orbits. In a chaotic region, trajectories are not confined in this manner and can explore a much larger part of the available phase space.

For nonlinear Hamiltonian systems, the transition between regular and chaotic behaviour is often not complete. Instead, regular and chaotic regions can coexist in the same phase space, producing what is known as a mixed phase space. The Kolmogorov-Arnold-Moser (KAM) theorem provides an important framework for understanding this structure. It shows that, under sufficiently small perturbations of an integrable Hamiltonian system, many of the invariant tori of the integrable system survive, although some are destroyed, particularly near resonances. The surviving invariant tori form regular regions of phase space, while the regions between them can develop increasingly complicated and chaotic dynamics. [3]

This mixed phase space containing regular and chaotic motion is particularly important for the kicked top studied in this work. Its classical phase space contains regular islands surrounded by a chaotic sea. The different regions therefore provide a natural setting for studying how quantum dynamics relates to different types of classical motion.

1.2 Quantum Chaos

The idea of chaos does not carry over directly from classical mechanics to quantum mechanics. In classical systems, chaos is associated with the divergence of nearby trajectories, but in quantum mechanics a state does not follow a definite trajectory in phase space. In addition, unitary quantum evolution preserves the overlap between two states. Therefore, the classical definition of chaos based on sensitive dependence on initial conditions cannot be applied directly to a quantum system [3].

This leads to the question of how the chaotic behaviour of a classical sys-

tem is reflected in its quantum counterpart. The connection between the two is studied through the quantum–classical correspondence. In the semiclassical limit, quantum dynamics can show features related to the underlying classical phase-space structure. Since quantum states cannot be represented as points in phase space, phase-space distributions such as the Husimi function provide a useful way to visualize this correspondence.

One convenient class of systems for studying quantum chaos is given by **quantum maps**. A quantum map is the quantum counterpart of a classical discrete map and is represented by a unitary operator acting once during each iteration. Periodically driven systems naturally lead to such maps through their Floquet operators [3]. The kicked top is one such system, and its classical dynamics can contain both regular islands and a chaotic sea. Studying the quantum evolution of states initially placed in these different regions therefore provides a way to investigate the quantum signatures of classical chaos.

1.3 Floquet Systems

For a time-independent Hamiltonian, the stationary states of a quantum system are obtained from the time-independent Schrödinger equation

$$H\left|\psi_n\right\rangle = E_n\left|\psi_n\right\rangle,$$

and their time evolution only introduces a phase factor,

$$\left|\psi_n(t)\right\rangle = e^{-iE_nt/\hbar}\left|\psi_n\right\rangle.$$

However, for a time-dependent Hamiltonian $H(t)$, the Hamiltonian itself changes with time and the usual stationary-state description in terms of energy eigenstates is no longer applicable. This is the situation encountered in periodically driven systems such as the kicked top.

If the Hamiltonian is periodic in time with period T,

$$H(t+T) = H(t),$$

the evolution over one complete period can be described by a single unitary operator called the **Floquet operator**:

$$U_F = \mathcal{T}\exp\left[-\frac{i}{\hbar}\int_0^T H(t)\,dt\right],$$

where $\mathcal{T}$ denotes time ordering. It propagates the state from one period to the next,

$$\left|\psi(T)\right\rangle = U_F\left|\psi(0)\right\rangle.$$

After n periods, the evolution can therefore be written simply as

$$|\psi(nT)\rangle = U_F^n |\psi(0)\rangle .$$

Thus, although the Hamiltonian is explicitly time dependent, the same operator U_F is applied at every period, making the stroboscopic evolution a discrete-time problem [3].

Since the time evolution is unitary, the Floquet operator satisfies

$$U_F^\dagger U_F = I.$$

Its eigenstates are defined by

$$U_F |e_n\rangle = e^{-i\phi_n} |e_n\rangle ,$$

where $|e_n\rangle$ are the Floquet eigenstates and ϕ_n are the corresponding **eigenphases**. Because U_F is unitary, its eigenvalues have unit magnitude and therefore lie on the unit circle in the complex plane. The eigenphases are consequently real and are defined modulo 2π.

The Floquet eigenstates play a role similar to stationary states for a time-independent Hamiltonian. If the system is initially in one of these states, its evolution is

$$U_F^n |e_j\rangle = e^{-in\phi_j} |e_j\rangle ,$$

so that its physical state remains unchanged apart from an overall phase. For a general initial state written as

$$|\psi(0)\rangle = \sum_j c_j |e_j\rangle ,$$

the evolution is

$$|\psi(nT)\rangle = \sum_j c_j e^{-in\phi_j} |e_j\rangle .$$

Thus, the eigenphases determine how the different Floquet components accumulate relative phases during the evolution. This makes the Floquet eigenstates and eigenphases particularly useful for understanding the long-time dynamics of periodically driven quantum systems [3].

1.4 Kicked systems

1.4.1 Classical Kicked systems

A useful class of periodically driven systems is given by kicked Hamiltonians, in which the system receives short impulses at regular intervals of time. A

general classical kicked Hamiltonian can be written as

$$H(q,p,t) = f(p) + V(q) \sum_{n=-\infty}^{\infty} \delta\left(\frac{t}{T} - n\right),$$

where T is the time between two successive kicks. The delta function represents an idealized instantaneous kick. Between two kicks, the system evolves only under the time-independent part $f(p)$, while at the instant of the kick the potential $V(q)$ gives an instantaneous change in the momentum [3].

Instead of following the continuous evolution at every instant of time, it is convenient to observe the system immediately after each kick. This gives a discrete map, called the **stroboscopic map**. For the Hamiltonian above, the equations of motion are

$$\frac{dq}{dt} = f'(p), \qquad \frac{dp}{dt} = -\delta_T(t)V'(q),$$

where $\delta_T(t)$ represents the periodic train of delta functions. During the time between two kicks, the momentum remains unchanged and the position evolves freely. At the next kick, the position is essentially unchanged during the infinitesimal kick interval, while the momentum receives an instantaneous impulse. Combining these two parts gives the stroboscopic map

$$q_{n+1} = q_n + Tf'(p_n),$$

$$p_{n+1} = p_n - TV'(q_{n+1}).$$

Thus, a continuous-time periodically kicked Hamiltonian can be reduced to a discrete-time map by considering the system at the same point in each driving cycle [3].

The stroboscopic map is closely related to the **Poincaré surface of section (PSOS)**. In a Poincaré section, a continuous trajectory is recorded whenever it crosses a chosen surface in phase space. The stroboscopic map follows a similar idea, but the points are recorded at fixed time intervals equal to the driving period T. Therefore, the return time is fixed for a stroboscopic map, whereas the time between successive crossings of a general Poincaré section need not be fixed. Despite this difference, both representations reduce the continuous dynamics to a discrete map and reveal the underlying regular and chaotic structures of the phase space [3].

1.4.2 Quantum Kicked systems

The same idea can be extended to quantum mechanics. The quantum counterpart of a classical map is a unitary operator that connects the quantum

state immediately after one kick to the state immediately after the next kick. Consider the quantum kicked Hamiltonian

$$\hat{H}(t) = \hat{H}_0 + \hat{V} \sum_{n=-\infty}^{\infty} \delta\left(\frac{t}{T} - n\right).$$

The quantum state just after the nth kick can be written as $|\psi(n^+)\rangle$. The quantum map, or Floquet operator, is then defined by

$$\left|\psi((n+1)^+)\right\rangle = U_F \left|\psi(n^+)\right\rangle.$$

Since the Hamiltonian is periodic, the same operator U_F is applied after every kick. The operator is unitary because it represents quantum time evolution, and therefore

$$U_F^\dagger U_F = I.$$

For an ideal instantaneous kick, the evolution over one period consists of free evolution under $\hat{H}_0$ and the instantaneous evolution generated by $\hat{V}$.

1.4.3 Derivation of Floquet operator

The time-evolution operator generated by a time-dependent Hamiltonian $H(t)$ is formally given by

$$\mathcal{U}(t_2, t_1) = \mathcal{T} \exp\left[-\frac{i}{\hbar}\int_{t_1}^{t_2} H(t')\, dt'\right], \tag{1.1}$$

where $\mathcal{T}$ denotes the time-ordering operator. For a periodically driven system satisfying $H(t+T) = H(t)$, the evolution over one period is described by the Floquet operator.

Consider the periodically kicked Hamiltonian

$$H(t) = H_0 + V \sum_{n=-\infty}^{\infty} \delta(t - nT), \tag{1.2}$$

where H_0 is the time-independent evolution between kicks, V is the kick potential, and T is the kicking period. Choosing the stroboscopic times immediately after successive kicks, $t_n = nT$, the one-period evolution is

$$U_F = \mathcal{U}(t_{n+1} + \epsilon, t_n + \epsilon), \qquad \epsilon \to 0^+. \tag{1.3}$$

Using the composition property of the time-evolution operator, the evolution over one period can be separated into the kick and the free-evolution parts:

$$U_F = \mathcal{U}(t_{n+1} + \epsilon, t_{n+1} - \epsilon)\mathcal{U}(t_{n+1} - \epsilon, t_n + \epsilon). \tag{1.4}$$

Across the infinitesimally short kick, the kick potential dominates and

$$\begin{aligned}\mathcal{U}(t_{n+1}+\epsilon, t_{n+1}-\epsilon) &= \exp\left[-\frac{i}{\hbar}\int_{t_{n+1}-\epsilon}^{t_{n+1}+\epsilon} V\sum_k \delta(t'-kT)\,dt'\right] \\ &= \exp\left(-\frac{i}{\hbar}V\right), \end{aligned} \tag{1.5}$$

where time ordering is trivial since V is time independent.

Between two successive kicks, the delta-function terms vanish and $H(t) = H_0$. Therefore,

$$\mathcal{U}(t_{n+1}-\epsilon, t_n+\epsilon) = \exp\left(-\frac{i}{\hbar}H_0 T\right), \tag{1.6}$$

in the limit $\epsilon \to 0^+$. Hence, the Floquet operator in the post-kick stroboscopic frame is [3]

$$\boxed{U_F = \exp\left(-\frac{i}{\hbar}V\right)\exp\left(-\frac{i}{\hbar}H_0 T\right)}. \tag{1.7}$$

Kicked systems are particularly useful in the study of quantum chaos because the classical dynamics can be represented by a simple discrete map, while the quantum dynamics is represented by repeated application of a finite-dimensional unitary operator. This makes it possible to compare the classical phase-space structure directly with the evolution of quantum states. The kicked top studied in this work is an example of such a system.

Chapter 2

Theory and Methods

2.1 Quantum Kicked Top

2.1.1 Theoretical Hamiltonian

The kicked top is a paradigmatic model for studying quantum chaos. It describes a spin F subjected to periodic driving and provides a direct connection between quantum dynamics and the corresponding classical phase-space dynamics. The system is described by the time-periodic Hamiltonian [1]

$$H(t) = \hbar p F_y \sum_{n=0}^{\infty} f(t - n\tau) + \hbar \frac{k}{2F\tau} F_x^2, \tag{2.1}$$

where p determines the rotation about the y-axis, k controls the strength of the nonlinear interaction, and τ is the driving period. In the idealized delta-kick limit, each period consists of a rotation about the y-axis followed by a nonlinear twist about the x-axis, with the twist angle proportional to F_x. The nonlinear F_x^2 term is responsible for generating the complex phase-space structure, including regular islands and chaotic regions. [1]

In this work, the kicked top is used to investigate the correspondence between classical and quantum dynamics. The evolution of spin coherent states initially located in different regions of the classical phase space is studied using the Husimi representation.

2.1.2 Experimental Implementation

The quantum kicked top is experimentally implemented using an ensemble of approximately 10^7 cesium (^{133}Cs) atoms. The atoms are first laser-cooled in a magneto-optical trap followed by optical molasses, after which they are

released into free fall and optically pumped into the $F = 3$ ground-state hyperfine manifold. Spin coherent states along desired directions are then prepared using controlled magnetic-field pulses. [1]

The two main parts of the kicked-top dynamics are implemented using different interactions. The rotation of the spin is produced using magnetic-field pulses generated by precision coils driven by arbitrary waveform generators. The magnetic interaction is described by the Zeeman Hamiltonian

$$H_{\mathrm{Z}} = g_F \mu_B \, \mathbf{B} \cdot \mathbf{F}, \tag{2.2}$$

where g_F is the Landé g-factor and μ_B is the Bohr magneton. These pulses allow controlled rotations of the spin and form the rotational part of the kicked-top dynamics.

The nonlinear twisting interaction is generated using a linearly polarized, monochromatic laser field. The resulting AC Stark shift produces an effective interaction proportional to F_x^2,

$$H_{\mathrm{twist}} = \hbar \xi F_x^2, \tag{2.3}$$

where $\xi = 2\pi \times 533$ Hz determines the strength of the twisting interaction. Thus, the magnetic field provides the rotation, while the laser-induced AC Stark shift provides the nonlinear twisting required for the kicked top. [1]

For the parameters used in the experiment, the kicked-top period is $\tau = 100$ μs and the twisting strength corresponds to $k = 2.0$. Due to spatial inhomogeneity of the laser field, the value of the twisting parameter k varies by approximately 5% across the atomic ensemble. The rotation parameter is chosen as $p = 0.99$, giving a mixed classical phase space containing both regular and chaotic regions. The finite duration of the magnetic pulse also leads to a small overlap between the rotation and twisting parts of the evolution.

A small static magnetic field of approximately 2 mG is applied along the z-axis to reduce the effects of background magnetic-field fluctuations and to make the dynamical tunnelling between the regular islands more robust. The interaction associated with this bias field will be included in the Hamiltonian in the following subsection.

Decoherence in the experiment is mainly caused by spontaneous photon scattering from the laser field. The laser parameters are chosen to keep the scattering rate sufficiently low so that the dynamics remain predominantly coherent over the timescale of the experiment. This experimental implementation therefore provides independent control over the linear rotation and nonlinear twisting terms, allowing the quantum dynamics of the kicked top to be observed over several kicks.

2.1.3 Experimental Hamiltonian

The ideal kicked-top Hamiltonian consists of a nonlinear twisting term and a periodic rotation term. However, the experimental system also contains a small static magnetic field applied along the z-axis (The paper does not explicitly specify the sign convention for the direction of the applied bias field.). The original paper states that a DC bias field of approximately 2 mG was applied and included in the theoretical modelling. The supplementary material also shows that this bias field was important for obtaining robust dynamical tunnelling. However, the explicit Hamiltonian term corresponding to this additional bias field is not given in either the main paper or its supplementary material [1].

The additional interaction can be obtained from the magnetic dipole interaction between the atomic spin and the applied magnetic field. The magnetic interaction is given by

$$H_Z = -\boldsymbol{\mu} \cdot \mathbf{B}, \tag{2.4}$$

where $\boldsymbol{\mu}$ is the magnetic moment of the hyperfine spin and $\mathbf{B}$ is the applied magnetic field. For a hyperfine state with total angular momentum F, the magnetic moment is

$$\boldsymbol{\mu} = -g_F \frac{\mu_B}{\hbar} \mathbf{F}, \tag{2.5}$$

where g_F is the hyperfine Landé g-factor, μ_B is the Bohr magneton, and $\mathbf{F}$ denotes the angular momentum operator. Substituting this into the magnetic interaction gives

$$H_Z = g_F \frac{\mu_B}{\hbar} \mathbf{F} \cdot \mathbf{B}. \tag{2.6}$$

For the bias field, assuming it's applied along the positive z-axis,

$$\mathbf{B} = B_z \hat{z},$$

the interaction reduces to

$$H_Z = g_F \frac{\mu_B B_z}{\hbar} F_z. \tag{2.7}$$

It is convenient to express this term in the dimensionless form used for one period of the kicked-top evolution,

$$H_Z = \frac{\hbar a}{\tau} F_z, \tag{2.8}$$

where

$$a = \frac{g_F \mu_B B_z \tau}{\hbar}. \tag{2.9}$$

For the $F = 3$ ground-state hyperfine manifold of ^{133}Cs, $g_F \simeq -0.25$. Taking the experimentally reported bias-field magnitude $B_z = 2$ mG $= 2 \times 10^{-7}$ T and the kicked-top period $\tau = 100$ μs gives

$$a = \frac{(-0.25)(9.274 \times 10^{-24})(2 \times 10^{-7})(100 \times 10^{-6})}{1.055 \times 10^{-34}} \simeq -0.44.$$

The bias field is assumed initially to be directed along the positive z-axis, i.e. $\mathbf{B} = B_z \hat{z}$ with $B_z = 2$ mG. Using the values $g_F \simeq -0.25$, $\mu_B = 9.274 \times 10^{-24}$ J/T, $\tau = 100$ μs, and $\hbar = 1.055 \times 10^{-34}$ J s, the dimensionless bias-field parameter is estimated as

$$a = \frac{g_F \mu_B B_z \tau}{\hbar} \simeq -0.44.$$

However, the numerical model gives the closest agreement with the Floquet eigenphases reported in the paper for

$$\boxed{a = +0.4576}.$$

The difference in sign can be understood by changing the convention for the direction of the bias field. If the field is instead taken to be directed along the negative z-axis, $B_z \to -B_z$, the sign of a is reversed, giving a positive value. Thus, the positive value obtained numerically is consistent with choosing the experimental bias field to point along the negative z-direction.

reverses the sign of a.

The magnitude of the numerically obtained value is close to the analytical estimate, with

$$\frac{|0.4576| - |0.44|}{|0.44|} \times 100 \approx 4\%.$$

This small difference is expected since the analytical estimate is based on the reported value of bias-field strength of 2 mG,and it doesn't account for experimental variations . The value $a = 0.4576$ was therefore used in the numerical simulations, as it reproduces the published Floquet eigenphases most closely.

Including this additional bias-field interaction, the complete Hamiltonian used in the numerical modelling is

$$\boxed{H(t) = \hbar p F_y \sum_{n=-\infty}^{\infty} f(t - n\tau) + \hbar \frac{k}{2F\tau} F_x^2 + \frac{\hbar a}{\tau} F_z} \tag{2.10}$$

where the first term describes the magnetic-field-induced rotation, the second term represents the nonlinear twisting interaction generated by the AC Stark shift, and the final term accounts for the experimentally applied static bias field.

2.1.4 Floquet Operator

Comparing the experimental Hamiltonian in Eq. (2.10) with the general form of the periodically kicked Hamiltonian in Eq. (1.7), and taking $f(t - n\tau)$ to be a delta function, the Hamiltonian can be written in the form

$$H(t) = H_0 + V \sum_n \delta(t - n\tau), \tag{2.11}$$

where the time-independent part and the kick potential are identified as

$$H_0 = \frac{\hbar k}{2F\tau} F_x^2 + \frac{\hbar a}{\tau} F_z, \tag{2.12}$$

and

$$V = \hbar p F_y. \tag{2.13}$$

Therefore, using the general Floquet operator for kicked systems derived earlier,

$$U_F = \exp\left(-\frac{i}{\hbar} V\right) \exp\left(-\frac{i}{\hbar} H_0 \tau\right), \tag{2.14}$$

the Floquet operator for the experimental kicked top becomes

$$\boxed{U_F = \exp\left[-ipF_y\right] \exp\left[-i\left(\frac{k}{2F} F_x^2 + aF_z\right)\right].} \tag{2.15}$$

Here, the first exponential represents the rotation generated by the periodic magnetic-field pulse, while the second exponential contains the nonlinear twisting interaction and the additional static bias-field interaction. This Floquet operator is used for the subsequent numerical evolution and Floquet eigenstate analysis.

2.2 Classical Map

2.2.1 Derivation of the Classical Map

General Semiclassical Procedure

The classical map corresponding to a quantum Floquet operator can be obtained by considering the evolution of the dynamical operators in the Heisenberg picture and then taking the semiclassical limit. The procedure consists of three main steps.

First, for a periodically driven quantum system, the Floquet operator $\mathcal{U}$ describes the evolution over one driving period. If $\hat{\mathbf{A}}$ represents a set

of dynamical observables, its evolution over one period in the Heisenberg picture is given by [4]

$$\hat{\mathbf{A}}' = \mathcal{U}^\dagger \hat{\mathbf{A}} \mathcal{U}. \tag{2.16}$$

If the Floquet operator consists of a sequence of different operations, these operations can be applied successively. For example, if

$$\mathcal{U} = \mathcal{U}_1 \mathcal{U}_2, \tag{2.17}$$

then

$$\hat{\mathbf{A}}' = \mathcal{U}_2^\dagger \mathcal{U}_1^\dagger \hat{\mathbf{A}} \mathcal{U}_1 \mathcal{U}_2. \tag{2.18}$$

Thus, the transformations of the operators can be evaluated one at a time.

Second, the individual operator transformations are evaluated using the commutation relations of the system. For continuous transformations, this can be done using the Hadamard lemma,

$$e^{\hat{A}} \hat{B} e^{-\hat{A}} = \hat{B} + [\hat{A}, \hat{B}] + \frac{1}{2!}[\hat{A}, [\hat{A}, \hat{B}]] + \cdots . \tag{2.19}$$

For spin systems, the angular momentum commutation relations and the ladder operators provide a convenient way of evaluating these transformations.

Finally, the quantum operators are related to classical phase-space variables by taking the semiclassical limit. For a spin of quantum number F, we introduce the normalized variables

$$X = \frac{\hat{F}_x}{F}, \qquad Y = \frac{\hat{F}_y}{F}, \qquad Z = \frac{\hat{F}_z}{F}. \tag{2.20}$$

In the limit $F \to \infty$, these normalized operators behave as classical variables. Since the magnitude of the angular momentum is fixed, the corresponding classical variables satisfy

$$X^2 + Y^2 + Z^2 = 1. \tag{2.21}$$

The quantum commutators also reduce to the corresponding classical Poisson brackets in this limit. Therefore, the quantum evolution of the operators can be converted into a classical evolution of the phase-space coordinates. This gives a discrete classical map of the form

$$\mathbf{X}_{n+1} = M(\mathbf{X}_n), \tag{2.22}$$

where $\mathbf{X} = (X, Y, Z)$ represents a point on the classical phase space of the spin.

Classical Map for the Kicked Top

The Floquet operator considered in this work is

$$U_F = \exp\Big(-ip\hat{F}_y\Big) \exp\left[-i\left(\frac{k}{2F}\hat{F}_x^2 + a\hat{F}_z\right)\right],$$

where p is the linear rotation parameter, k is the nonlinear kicking strength, and a represents an additional rotation about the z-axis.

To obtain the classical map, we first consider the evolution of the angular momentum operator

$$\hat{\mathbf{F}} = (\hat{F}_x, \hat{F}_y, \hat{F}_z)$$

in the Heisenberg picture. After one period,

$$\hat{\mathbf{F}}' = U_F^\dagger \hat{\mathbf{F}} U_F.$$

Substituting the Floquet operator gives

$$\hat{\mathbf{F}}' = U_{\rm nl}^\dagger U_l^\dagger \hat{\mathbf{F}} U_l U_{\rm nl}, \tag{2.23}$$

where

$$U_l = \exp\Big(-ip\hat{F}_y\Big) \tag{2.24}$$

is the linear rotation and

$$U_{\rm nl} = \exp\left[-i\left(\frac{k}{2F}\hat{F}_x^2 + a\hat{F}_z\right)\right] \tag{2.25}$$

is the nonlinear part.

This expression shows clearly that the two operations act successively. Although the rightmost operator acts first on the state in the Schrödinger picture, the corresponding Heisenberg transformation of the observables is evaluated in the reverse operator order. The angular momentum vector is first transformed by the linear rotation, giving an intermediate vector $\hat{\mathbf{F}}''$,

$$\hat{\mathbf{F}}'' = U_l^\dagger \hat{\mathbf{F}} U_l,$$

and this rotated vector subsequently undergoes the nonlinear evolution,

$$\hat{\mathbf{F}}' = U_{\rm nl}^\dagger \hat{\mathbf{F}}'' U_{\rm nl}.$$

Thus, one complete kick can be viewed as

$$\boxed{\hat{\mathbf{F}} \xrightarrow{U_l} \hat{\mathbf{F}}'' \xrightarrow{U_{\rm nl}} \hat{\mathbf{F}}'}.$$

Linear Rotation

The first operation is a rotation about the y-axis generated by

$$U_l = \exp\Big(-ip\hat{F}_y\Big).$$

The transformed angular momentum operator is therefore

$$\hat{\mathbf{F}}'' = e^{ip\hat{F}_y}\hat{\mathbf{F}}e^{-ip\hat{F}_y}.$$

To evaluate this transformation, we use the Hadamard lemma together with the angular momentum commutation relations

$$[\hat{F}_i, \hat{F}_j] = i\epsilon_{ijk}\hat{F}_k,$$

where $\hbar = 1$ is used. For the x component,

$$e^{ip\hat{F}_y}\hat{F}_x e^{-ip\hat{F}_y} = \hat{F}_x + ip[\hat{F}_y, \hat{F}_x] + \frac{(ip)^2}{2!}[\hat{F}_y, [\hat{F}_y, \hat{F}_x]] + \cdots . \tag{2.26}$$

Using

$$[\hat{F}_y, \hat{F}_x] = -i\hat{F}_z, \qquad [\hat{F}_y, \hat{F}_z] = i\hat{F}_x,$$

the series becomes

$$\begin{aligned}\hat{F}_x'' &= \hat{F}_x + p\hat{F}_z - \frac{p^2}{2!}\hat{F}_x - \frac{p^3}{3!}\hat{F}_z + \cdots \\ &= \hat{F}_x \cos p + \hat{F}_z \sin p. \end{aligned} \tag{2.27}$$

Similarly,

$$\hat{F}_y'' = \hat{F}_y, \tag{2.28}$$

$$\hat{F}_z'' = -\hat{F}_x \sin p + \hat{F}_z \cos p. \tag{2.29}$$

Therefore, after the linear rotation,

$$\hat{F}_x'' = \hat{F}_x \cos p + \hat{F}_z \sin p, \tag{2.30}$$

$$\hat{F}_y'' = \hat{F}_y, \tag{2.31}$$

$$\hat{F}_z'' = -\hat{F}_x \sin p + \hat{F}_z \cos p. \tag{2.32}$$

Introducing the normalized classical variables

$$X = \frac{F_x}{F}, \qquad Y = \frac{F_y}{F}, \qquad Z = \frac{F_z}{F},$$

the corresponding classical transformation is

$$X'' = X_n \cos p + Z_n \sin p, \tag{2.33}$$
$$Y'' = Y_n, \tag{2.34}$$
$$Z'' = -X_n \sin p + Z_n \cos p. \tag{2.35}$$

Thus, the first part of the classical map is simply a rigid rotation of the spin vector about the y-axis.

Nonlinear Twist

The second operation is generated by the nonlinear part of the Floquet operator,

$$U_{\rm nl} = \exp\left[-i\left(\frac{k}{2F}\hat{F}_x^2 + a\hat{F}_z\right)\right]. \tag{2.36}$$

To obtain the corresponding classical evolution, it is useful to rewrite this operator in terms of the normalized spin variables. Since

$$\hat{X} = \frac{\hat{F}_x}{F}, \qquad \hat{Z} = \frac{\hat{F}_z}{F}, \tag{2.37}$$

we have

$$\frac{k}{2F}\hat{F}_x^2 + a\hat{F}_z = F\left(\frac{k}{2}\hat{X}^2 + a\hat{Z}\right). \tag{2.38}$$

Therefore,

$$U_{\rm nl} = \exp\left[-iF\left(\frac{k}{2}\hat{X}^2 + a\hat{Z}\right)\right]. \tag{2.39}$$

In the classical limit, the operators $\hat{X}$ and $\hat{Z}$ are replaced by the corresponding classical coordinates X and Z. The quantity inside the brackets therefore becomes the dimensionless classical Hamiltonian associated with the nonlinear part,

$$H_{\rm nl}(X, Y, Z) = \frac{k}{2}X^2 + aZ. \tag{2.40}$$

This conversion from the quantum Hamiltonian to a classical Hamiltonian is necessary because the classical dynamics are no longer described by operator evolution. Instead, the phase-space coordinates evolve according to Hamilton's equations, or equivalently, through their Poisson brackets with the classical Hamiltonian.

For the spin coordinates on the unit sphere, the Poisson brackets are

$$\{X_i, X_j\} = \epsilon_{ijk} X_k. \tag{2.41}$$

Hence, the evolution of any phase-space coordinate X_i is given by

$$\frac{dX_i}{d\tau} = \{X_i, H_{\rm nl}\}, \tag{2.42}$$

where τ is a dimensionless time variable.

Using

$$H_{\rm nl} = \frac{k}{2}X^2 + aZ, \tag{2.43}$$

we obtain

$$\frac{dX}{d\tau} = \left\{X, \frac{k}{2}X^2 + aZ\right\} = a\{X, Z\} = -aY, \tag{2.44}$$

$$\frac{dY}{d\tau} = \left\{Y, \frac{k}{2}X^2 + aZ\right\} = kX\{Y, X\} + a\{Y, Z\} = (a - kZ)X, \tag{2.45}$$

$$\frac{dZ}{d\tau} = \left\{Z, \frac{k}{2}X^2 + aZ\right\} = kX\{Z, X\} = kXY. \tag{2.46}$$

Therefore, the nonlinear part of the classical dynamics is described by the coupled equations

$$\boxed{\begin{aligned} \frac{dX}{d\tau} &= -aY, \\ \frac{dY}{d\tau} &= (a - kZ)X, \\ \frac{dZ}{d\tau} &= kXY. \end{aligned}} \tag{2.47}$$

These equations describe a continuous nonlinear evolution of the spin vector. The evolution preserves the constraint

$$X^2 + Y^2 + Z^2 = 1, \tag{2.48}$$

so that the trajectory remains on the unit sphere.

Complete Classical Map

The two parts of the Floquet operator now give the complete classical evolution during one kick. Starting from the point (X_n, Y_n, Z_n), the linear rotation is first applied to obtain (X'', Y'', Z''). This point is then used as the initial condition for the nonlinear equations of motion,

$$\mathbf{X}(0) = (X'', Y'', Z''). \tag{2.49}$$

The nonlinear equations are integrated over one driving period. In terms of the dimensionless time τ, this corresponds to

$$0 \leq \tau \leq T, \tag{2.50}$$

where T is the period of the kick. With the time scaled such that one driving period corresponds to $\tau = 1$, the final point is

$$\mathbf{X}_{n+1} = \mathbf{X}(\tau = 1). \tag{2.51}$$

Thus, one complete iteration of the classical map can be summarized as

$$\boxed{(X_n, Y_n, Z_n) \xrightarrow{\text{linear rotation}} (X'', Y'', Z'') \xrightarrow{\text{nonlinear twist}} (X_{n+1}, Y_{n+1}, Z_{n+1})}.$$

Repeating this procedure for many initial conditions generates the classical phase-space portrait of the kicked top. The resulting portrait reveals the regular islands and chaotic regions of the dynamics, which are later used to select the initial conditions for the corresponding quantum evolution.

2.3 Phase Space Representation of Quantum systems

In classical mechanics, the state of a system can be represented by a point in phase space. For example, the state of a particle is specified by its position and momentum (q, p), and its time evolution corresponds to a trajectory through phase space. This provides a direct way of visualizing the dynamics of a classical system. However, a similar description is not directly possible for a quantum system because position and momentum cannot be simultaneously specified with complete precision. They satisfy the Heisenberg uncertainty relation

$$\Delta q \, \Delta p \geq \frac{\hbar}{2}. \tag{2.52}$$

Therefore, a quantum state cannot be associated with a single point (q, p) in phase space in the same way as a classical state[5]. This difference becomes particularly important when studying the quantum analogue of classical dynamical systems. Since quantum mechanics does not provide well-defined trajectories in phase space, the classical picture of regular and chaotic trajectories cannot be directly applied to a quantum state. It is therefore useful to introduce a state in the Hilbert space which is the closest analogue of a classical phase space point. These states are called coherent states. [6]

2.3.1 Spin Coherent States

Classical and Quantum Angular Momentum

In classical mechanics, an angular momentum vector $\mathbf{L}$ has a fixed magnitude when there is no external torque. Therefore, it can be represented by a point on the surface of a sphere S^2. The direction of the vector can be specified by the polar and azimuthal angles (θ, ϕ), giving

$$\mathbf{L} = L \begin{pmatrix} \sin\theta\cos\phi \\ \sin\theta\sin\phi \\ \cos\theta \end{pmatrix}. \tag{2.53}$$

Thus, the classical phase space of a spin with fixed magnitude is the surface of a sphere.

In quantum mechanics, the components of angular momentum are represented by operators $\hat{J}_x$, $\hat{J}_y$ and $\hat{J}_z$, which satisfy

$$[\hat{J}_i, \hat{J}_j] = i\hbar\epsilon_{ijk}\hat{J}_k. \tag{2.54}$$

The total angular momentum operator is

$$\hat{J}^2 = \hat{J}_x^2 + \hat{J}_y^2 + \hat{J}_z^2, \tag{2.55}$$

and its eigenstates can be written as

$$\hat{J}^2|j,m\rangle = \hbar^2 j(j+1)|j,m\rangle, \tag{2.56}$$

$$\hat{J}_z|j,m\rangle = \hbar m|j,m\rangle, \tag{2.57}$$

where

$$m = -j, -j+1, \ldots, j. \tag{2.58}$$

For a fixed value of j, the Hilbert space therefore has dimension $2j+1$.

Unlike the classical angular momentum components, the quantum operators $\hat{J}_x$, $\hat{J}_y$ and $\hat{J}_z$ cannot be simultaneously measured with complete precision because they do not commute. Therefore, a quantum state cannot be represented by a sharply defined point on the classical sphere in the same way as a classical spin vector. Spin coherent states provide a useful way of establishing a correspondence between the quantum state and a point on this classical phase space.

Construction of Spin Coherent States

Classically, any point on the sphere can be reached by rotating a vector initially pointing along the z-axis. The same idea can be used in quantum mechanics. We start with the highest-weight state $|j, j\rangle$, which corresponds to a state with maximum projection along the z-axis. A spin coherent state pointing in the direction (θ, ϕ) is obtained by rotating this state:

$$|\theta, \phi\rangle = e^{-i\phi \hat{J}_z/\hbar} e^{-i\theta \hat{J}_y/\hbar} |j, j\rangle. \tag{2.59}$$

The expectation value of the angular momentum in this state points along the corresponding direction on the sphere,

$$\langle \theta, \phi | \hat{\mathbf{J}} | \theta, \phi \rangle = \hbar j \, \hat{\mathbf{n}}, \tag{2.60}$$

where

$$\hat{\mathbf{n}} = (\sin\theta\cos\phi, \, \sin\theta\sin\phi, \, \cos\theta). \tag{2.61}$$

Thus, each spin coherent state can be associated with a particular point (θ, ϕ) on the classical phase space.

The state $|j, j\rangle$ is also special because it has the minimum possible uncertainty in the two transverse angular momentum components. For this state,

$$\Delta J_x \, \Delta J_y = \frac{\hbar^2 j}{2}, \tag{2.62}$$

which satisfies the angular momentum uncertainty relation. Since the other coherent states are obtained from $|j, j\rangle$ by a rotation, they have the same uncertainty properties. Hence, spin coherent states are the most localized states that can be associated with a particular direction on the spin sphere.

Spin Coherent States in the $|j, m\rangle$ Basis

For numerical calculations, it is useful to express the coherent state in the standard angular momentum basis. We introduce the complex stereographic coordinate

$$z = \tan\left(\frac{\theta}{2}\right) e^{-i\phi}. \tag{2.63}$$

Using the angular momentum ladder operators

$$\hat{J}_\pm = \hat{J}_x \pm i\hat{J}_y, \tag{2.64}$$

the coherent state can be written as

$$|\theta, \phi\rangle = (1 + |z|^2)^{-j} e^{z\hat{J}_-/\hbar} |j, j\rangle. \tag{2.65}$$

Expanding this expression in the $|j,m\rangle$ basis gives

$$|\theta,\phi\rangle = (1+|z|^2)^{-j} \sum_{m=-j}^{j} \sqrt{\binom{2j}{j+m}}\, z^{j-m}|j,m\rangle. \tag{2.66}$$

This expression is the form used to construct the initial states in the numerical calculations. Although the states are labelled by the two parameters (θ,ϕ), they are not mutually orthogonal. Instead, the spin coherent states form an overcomplete set and satisfy the completeness relation

$$\frac{2j+1}{4\pi}\int d\Omega\, |\theta,\phi\rangle\langle\theta,\phi| = \hat{\mathbb{I}}, \tag{2.67}$$

where

$$d\Omega = \sin\theta\, d\theta\, d\phi. \tag{2.68}$$

Overlap and the Semiclassical Limit

The overlap between two spin coherent states gives a measure of how close their corresponding points are on the classical sphere. For two coherent states $|z\rangle$ and $|z'\rangle$, the overlap is

$$\langle z'|z\rangle = \frac{(1+z'^{*}z)^{2j}}{(1+|z'|^2)^j(1+|z|^2)^j}. \tag{2.69}$$

If the two states correspond to classical directions separated by an angle γ, the magnitude squared of the overlap can be written as

$$|\langle z'|z\rangle|^2 = \left(\cos\frac{\gamma}{2}\right)^{4j}. \tag{2.70}$$

When the two directions are the same, $\gamma = 0$, the overlap is one. As the angular separation increases, the overlap decreases. In particular, for opposite points on the sphere, $\gamma = \pi$, the overlap vanishes. More importantly, as j increases, the overlap becomes increasingly localized around $\gamma = 0$. Therefore, in the semiclassical limit $j \to \infty$, coherent states associated with different points on the classical sphere become effectively orthogonal.

This property is important for connecting the quantum and classical descriptions. A spin coherent state becomes increasingly localized around a particular point of the classical phase space as j becomes large. Therefore, coherent states provide a natural choice of initial states when studying the quantum-classical correspondence in the kicked top.

2.3.2 Phase-Space Representation: The Husimi Function

Although a general quantum state can be written in the $|j, m\rangle$ basis as

$$|\psi\rangle = \sum_{m=-j}^{j} c_m |j, m\rangle, \tag{2.71}$$

this representation does not directly show where the state is localized on the classical phase space. To visualize the quantum state on the spin sphere, we use the spin coherent states introduced above.

The Husimi function gives the overlap of a quantum state with a coherent state at each point (θ, ϕ) of the sphere. For a density matrix ρ, it is defined as

$$H(\theta, \phi) = \frac{2j+1}{4\pi} \langle \theta, \phi | \rho | \theta, \phi \rangle. \tag{2.72}$$

For a pure state

$$\rho = |\psi\rangle\langle\psi|, \tag{2.73}$$

this reduces to

$$H(\theta, \phi) = \frac{2j+1}{4\pi} |\langle \theta, \phi | \psi \rangle|^2. \tag{2.74}$$

Thus, for a pure state, the Husimi function is basically the overlap of the wavefunction of the system with the spin coherent state. Since each spin coherent corresponds to a classical phase space point, we can obtain a quasi-probability distribution of the quantum system in the phase space . It is therefore very useful for comparing quantum dynamics with the classical phase-space structure of the kicked top. In this work, the Husimi function is used to visualize the evolution of spin coherent states initially placed in the regular and chaotic regions of the classical phase space.

2.3.3 Fidelity

Fidelity is a measure of the similarity between two quantum states. It is very useful for studying how much an evolved state differs from its initial state. For two general density matrices ρ_0 and ρ_n, the fidelity is defined as [1]

$$\mathcal{F}(\rho_0, \rho_n) = \left[\mathrm{Tr} \sqrt{\sqrt{\rho_0} \rho_n \sqrt{\rho_0}} \right]^2.$$

In the present work, the initial states are pure spin coherent states and the evolution is unitary. Therefore,

$$\rho_0 = |\psi_0\rangle \langle\psi_0|, \qquad \rho_n = |\psi_n\rangle \langle\psi_n|,$$

where

$$|\psi_n\rangle = U_F^n |\psi_0\rangle .$$

For pure states, the above expression reduces simply to the squared overlap between the two states,

$$\mathcal{F}_n = |\langle\psi_0|\psi_n\rangle|^2 .$$

Thus, in our case, the fidelity measures how closely the state after n kicks resembles the initial coherent state. A fidelity close to 1 means that the evolved state has significant overlap with the initial state, whereas a value close to 0 indicates that the state has evolved to a state nearly orthogonal to the initial state.

2.4 Numerical Modeling of the Schrodinger Evolution

2.4.1 Computing the Floquet Operator

Although we have an analytical expression for the Floquet operator, its better to compute it numerically solving the time-dependent Schrödinger equation by modeling the kick using a gaussian pulse over one driving period, since this approach is a little closer to experimental implementation of kick by applying a finite width magnetic field pulses. In the Schrödinger picture, the time evolution operator satisfies

$$\frac{dU(t)}{dt} = -iH(t)U(t), \tag{2.75}$$

where $\hbar = 1$ is used. The initial condition is

$$U(0) = \mathbb{I}. \tag{2.76}$$

Solving this equation from the beginning to the end of one driving period gives the corresponding time-evolution operator, which can then be used as the Floquet operator.

Numerical Representation of the Delta Kick

The kicked-top Hamiltonian contains an idealized delta-function kick. A delta function is difficult to use directly in a numerical integration. Therefore, in the numerical calculation, the delta kick is represented by a narrow Gaussian pulse. The pulse is defined as

$$f(t) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{t-t_c}{\sigma}\right)^2\right], \tag{2.77}$$

where the centre of the pulse is chosen as

$$t_c = \tau - 6\sigma. \tag{2.78}$$

Thus, the Gaussian is located very close to the end of the driving period. Since the Gaussian is normalized,

$$\int_{-\infty}^{\infty} f(t)\, dt = 1, \tag{2.79}$$

it approaches the effect of a delta function when σ is made sufficiently small.

In the numerical calculation, the Gaussian pulse is implemented as

$$f(t, \tau, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{t - (\tau - 6\sigma)}{\sigma}\right)^2\right]. \tag{2.80}$$

The use of $t_c = \tau - 6\sigma$ places the pulse almost entirely within the last part of the driving period without any boundary value error. This allows the kick to be treated numerically as a finite-width pulse while still closely approximating the ideal delta kick.

Time-Dependent Hamiltonian

Using the Gaussian representation of the kick, the Hamiltonian used for the numerical evolution is

$$H(t) = pf(t)\hat{F}_y + \frac{k}{2j\tau}\hat{F}_x^2 + \frac{a}{\tau}\hat{F}_z.$$

We have substituted $\hbar = 1$. The first term represents the short linear kick about the y-axis, while the remaining two terms describe the continuous part of the Hamiltonian. The finite-width Gaussian pulse is chosen to occur near the end of the period. Therefore, during most of the evolution the system evolves under the continuous terms, followed by the short kick near the end of the period. This reproduces the ordering of the operations in the Floquet description.

The time-evolution operator is obtained by integrating equation 2.75 by substituting $f(t, \tau, \sigma)$ into $H(t)$ from $t = 0$ to $t = \tau$, with $U(0) = \mathbb{I}$. The resulting operator at the end of the period,

$$U_F = U(\tau), \tag{2.81}$$

is therefore the numerical Floquet operator for one kick.

Numerical Integration

The matrix differential equation is solved using the `solve_ivp` routine from the `SciPy` library. The integration uses the fourth-order Runge–Kutta method with adaptive step size (RK45). The adaptive step size is particularly useful here because the Hamiltonian changes rapidly during the narrow Gaussian pulse. The solver can therefore take larger steps when the Hamiltonian varies slowly and automatically reduces the step size around the kick.

The parameters used for the numerical evolution are

$$a = 0.4576, p = 0.99, k_0 = 2.0, \tau = 1.0, \sigma = 0.001.$$

The value of σ is much smaller than the driving period, making the Gaussian pulse sufficiently narrow to approximate the delta-function kick.

For the simulations involving multiple kicks, the evolution operator for one period is used repeatedly to evolve the initial state. If $|\psi_0\rangle$ is the initial state, the state after n kicks is given by

$$|\psi_n\rangle = U_F^n |\psi_0\rangle. \tag{2.82}$$

The resulting state at each kick is then used to calculate the Husimi function and the fidelity.

Averaging over the Kicking Strength

To account for a 5% variation in the kicking strength, the simulations were also performed for 21 values of k around the nominal value $k_0 = 2$. The width used for this variation was

$$\sigma_k = 0.05 k_0. \tag{2.83}$$

For $k_0 = 2$, this gives

$$\sigma_k = 0.1. \tag{2.84}$$

For each value of k sampled from the uniform distribution, the time-dependent Schrödinger equation was solved independently and the corresponding quantum observables were calculated. The final results were then averaged over the different realizations. This averaging accounts for the small experimental variation in the kicking strength and gives a more realistic representation of the experimentally relevant dynamics. The decoherence was not modeled in this work.

Chapter 3

Results

3.1 Classical Dynamics

3.1.1 Classical Phase-Space Structure

The classical phase space of the kicked top is obtained by repeatedly applying the classical map derived from the Floquet operator. Since the magnitude of the classical angular momentum is conserved, the dynamics are confined to the surface of the unit sphere,

$$X^2 + Y^2 + Z^2 = 1.$$

To visualize the dynamics, the position of the spin vector is recorded after every kick. The resulting stroboscopic points are then projected onto the X-Z plane. The two hemispheres are plotted separately according to the sign of Y.

For the classical map considered in this work, the parameters are chosen as

$$p = 0.99, \qquad k = 2.0, \qquad a = 0.4576,$$

where p is the rotation angle about the y-axis, k is the nonlinear kicking strength, and a represents the additional term corresponding to the bias along the z-axis. The inclusion of the aZ term in the classical Hamiltonian modifies the phase-space structure compared with the case where this term is absent.

The resulting phase-space portrait is shown in Fig. 3.1. The phase space contains both regular and chaotic regions. Regular regions appear as well-defined closed structures, where neighbouring trajectories remain confined and follow ordered motion. In contrast, the chaotic sea covers regions where

the trajectories spread over a larger area and do not remain confined to a single regular structure.

In the $Y < 0$ hemisphere, a large regular island is clearly visible. In the $Y > 0$ hemisphere, a pair of smaller regular islands can be identified within the surrounding chaotic region. These structures provide different types of classical dynamics that can be used to study their quantum counterparts.

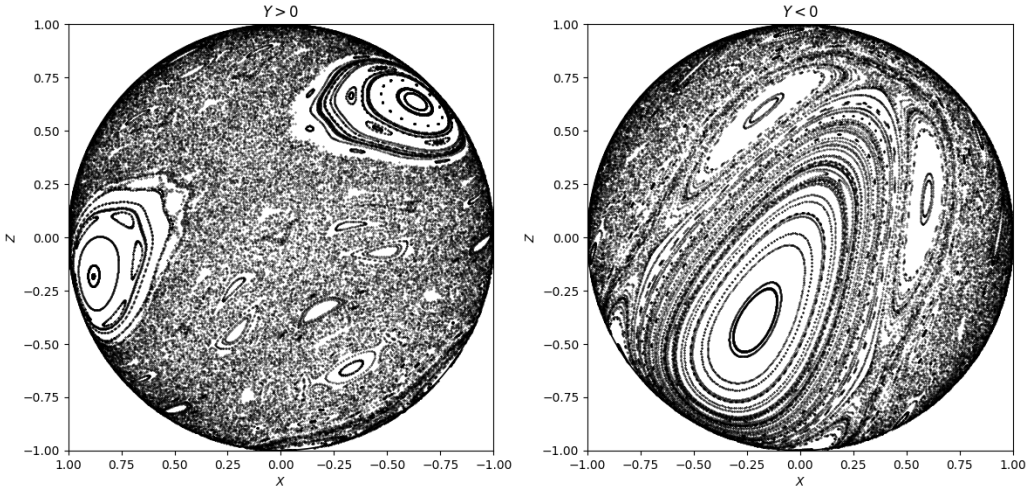


Figure 3.1: Classical phase-space portrait of the kicked top for $p = 0.99$, $k = 2.0$ and $a = 0.4576$. The trajectories are projected onto the X-Z plane and the two hemispheres are shown separately according to the sign of Y. A large regular island is present in the $Y < 0$ hemisphere, while smaller regular islands are present in the $Y > 0$ hemisphere, surrounded by a chaotic sea.

To study the quantum-classical correspondence in these different dynamical regions, three initial conditions are selected from the classical phase space. The first is chosen inside the large regular island, the second inside one of the smaller regular islands, and the third inside the chaotic sea. The corresponding initial conditions are

$$\text{Large regular island:} \quad (X, Y, Z) = (0.00,\ -0.99,\ -0.16), \tag{3.1}$$

$$\text{Small regular island:} \quad (X, Y, Z) = (0.70,\ 0.70,\ -0.16), \tag{3.2}$$

$$\text{Chaotic sea:} \quad (X, Y, Z) = (-0.94,\ 0.31,\ -0.16). \tag{3.3}$$

respectively.

These three initial conditions allow us to compare the evolution of quantum states initially localized in qualitatively different regions of the classical phase space.

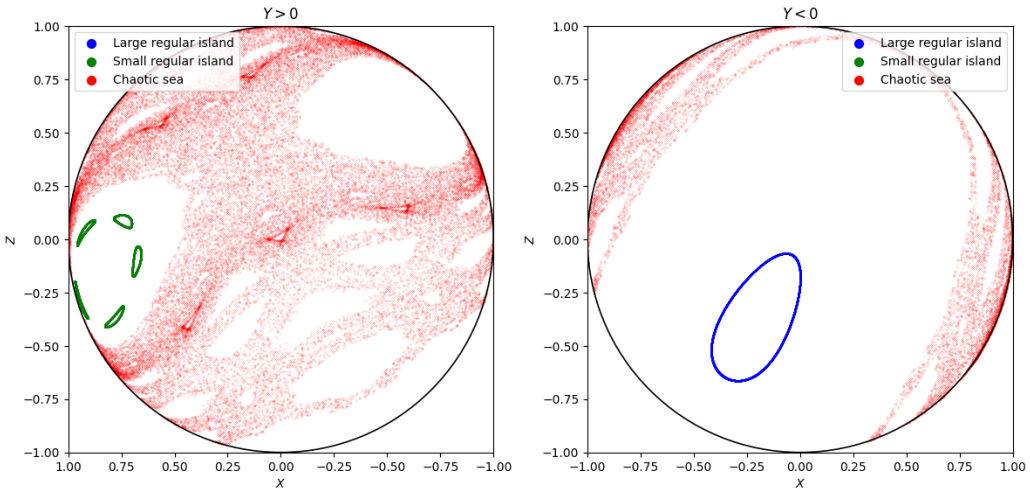


Figure 3.2: Classical phase-space evolution of the three initial condition for $p = 0.99$, $k = 2.0$ and $a = 0.4576$. The trajectories remain confined to the corresponding regular regions or spread through the chaotic sea according to the initial condition.

The three initial conditions correspond to qualitatively different classical dynamics. The initial condition chosen inside the large regular island, $(X, Y, Z) = (0.00, -0.99, -0.16)$, remains confined to a closed invariant curve, indicating regular and quasiperiodic motion. The initial condition $(X, Y, Z) = (0.70, 0.70, -0.16)$ lies in the smaller regular-island region. Its trajectory visits the different members of a five-period island chain and appears as a set of five regular structures in the stroboscopic phase space. In contrast, the initial condition $(X, Y, Z) = (-0.94, 0.31, -0.16)$ lies in the chaotic sea. Its trajectory spreads through the surrounding phase-space region and does not remain confined to a regular invariant curve, indicating chaotic dynamics. In the quantum description, the same points are used to construct spin coherent states, whose evolution is represented using the Husimi Q-function.

3.2 Quantum Dynamics

Having examined the classical phase-space structure of the kicked top, we now turn to its quantum dynamics. Throughout this section, the same system parameters as those used for the classical map are used.The objective is to investigate the quantum signatures corresponding to the different classical dynamical regions. In particular, we study the Floquet eigenstates, Floquet

eigenvalues (eigenphases), and the time evolution of quantum states initialized in the three representative regions identified in the classical phase space: the large regular island, the small regular island, and the chaotic sea. The quantum dynamics in phase space is analyzed using the Husimi Q-function, allowing a direct comparison between the classical and quantum descriptions

3.2.1 Floquet Eigenstates and Eigenphases

Since the Hamiltonian of the kicked system is explicitly time-dependent, its instantaneous eigenstates are generally not stationary states of the full time evolution. Instead, it is more convenient to consider the Floquet operator, which describes the evolution of the system over one complete driving period. The Floquet operator is time-independent and therefore its eigenstates provide stationary states with respect to the stroboscopic evolution.

The Floquet eigenstates $|e_i\rangle$ are defined by

$$U_F \left|e_i\right\rangle = e^{-i\Phi_i} \left|e_i\right\rangle ,$$

where ϕ_i is called the eigenphase. Since U_F is unitary, its eigenvalues have unit magnitude and therefore lie on the unit circle in the complex plane. They can consequently be written in the form

$$\lambda_i = e^{-i\Phi_i},$$

with the eigenphases Φ_i defined from $-\pi$ to π. The set of Floquet eigenstates and their corresponding eigenphases is therefore used to characterize the stationary structure of the periodically driven quantum system.

For the calculations in this work, we consider a spin quantum number $F = 3$. The corresponding Hilbert space has dimension

$$d = 2F + 1 = 7,$$

with the basis states $|F, m\rangle$, where $m = -F, -F + 1, \ldots, F$. Therefore, the 7×7 Floquet operator has seven eigenstates, each with a corresponding eigenphase. Thus, the Floquet spectrum of the system consists of seven eigenstate–eigenphase pairs,

$$\left\{\left|e_i\right\rangle , \Phi_i\right\} , \qquad i = 1, \ldots, 7.$$

The eigenphases can be obtained by numerically diagonalizing the Floquet operator calculated for one complete driving period. Once the eigenstates $|e_i\rangle$ are obtained, their distribution in the classical phase space can be visualized

using the Husimi Q-function. For a Floquet eigenstate $|e_i\rangle$, we define the Husimi function as

$$H_i(\theta, \phi) = \frac{2F+1}{4\pi} \left|\langle \theta, \phi | e_i \rangle\right|^2 ,$$

where $|\theta, \phi\rangle$ is the spin coherent state corresponding to the phase-space point (θ, ϕ). The Husimi function therefore shows where each Floquet eigenstate is localized in phase space and allows its structure to be compared directly with the classical phase-space portrait.

For consistency with the classical phase-space plots, the same projection and viewing convention is used here. In particular, the spin coherent states are parametrized by the same spherical coordinates (X, Y, Z), and the two hemispheres are separated according to the sign of Y. The Husimi distributions are consequently projected onto the X-Z plane in the same way as the classical trajectories. This allows the regular and chaotic structures observed in the classical phase space to be directly compared with the localization of the corresponding Floquet eigenstates.

The Husimi distributions of the Floquet eigenstates can be compared with the classical phase-space structure shown in Fig. 3.1. From the distributions in Fig. 3.3, it can be seen that the states $|e_1\rangle$ and $|e_4\rangle$ are mainly localized in the large regular island in the $Y < 0$ hemisphere. Similarly, the states $|e_2\rangle$ and $|e_3\rangle$ have significant overlap with the pair of smaller regular islands in the $Y > 0$ hemisphere. The remaining Floquet eigenstates are mostly distributed over the chaotic regions of the phase space, although some of them also show partial localization around the regular region. This correspondence between the Husimi distributions of the Floquet eigenstates and the underlying classical phase-space structure allows us to identify which eigenstates contribute most strongly to the dynamics of each of the three initial conditions.

3.2.2 Evolution of Spin Coherent States

As discussed earlier, three different initial conditions are chosen from the classical phase space in order to compare the quantum dynamics corresponding to regular and chaotic classical motion. To represent these classical points in the quantum system, we choose spin coherent states pointing in the corresponding directions (θ, ϕ) on the spin sphere. If the initial coherent state is denoted by $|\psi_0\rangle$, its evolution after n kicks is obtained by repeatedly applying the Floquet operator,

$$|\psi_n\rangle = U_F^n |\psi_0\rangle .$$

Here, U_F represents the evolution over one complete driving period. Therefore, the same initial phase-space points used in the classical simulation can

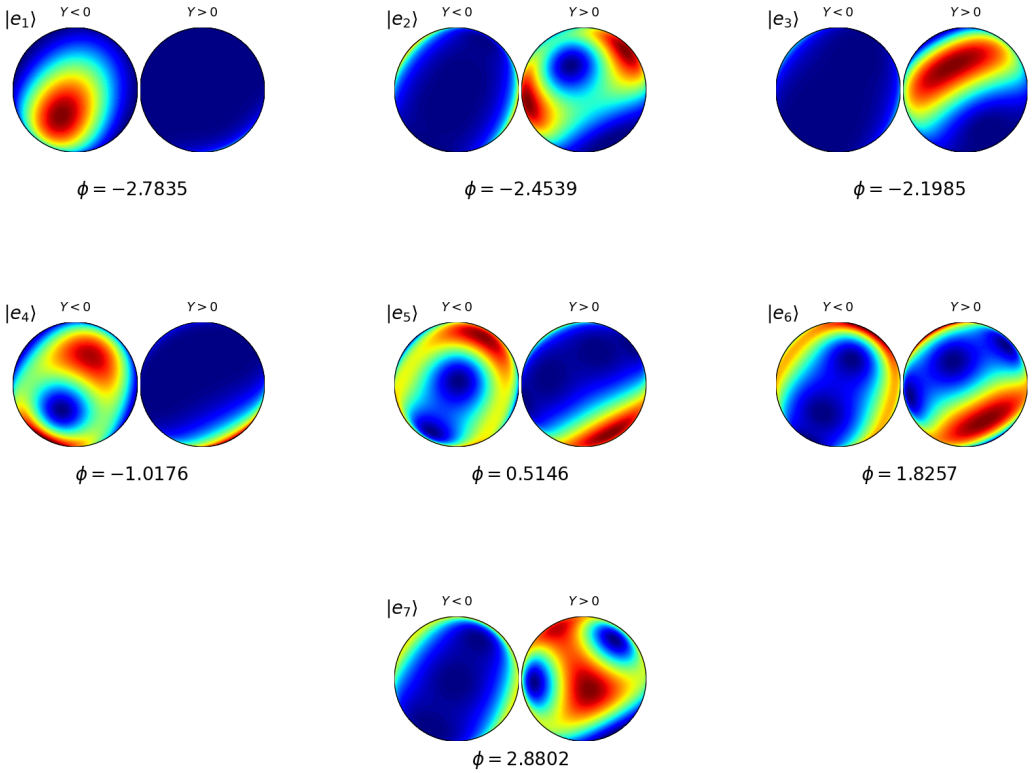


Figure 3.3: Husimi Q-function distributions of the seven Floquet eigenstates for $F = 3$, together with their corresponding eigenphases Φ_i.

be represented by quantum states and their evolution in the phase space can be studied.

To visualize the quantum evolution in phase space, the Husimi function is calculated for the state after each kick. For the state $|\psi_n\rangle = U_F^n |\psi_0\rangle$, the Husimi function is given by

$$H_n(\theta, \phi) = \frac{2F+1}{4\pi} |\langle \theta, \phi | U_F^n | \psi_0 \rangle|^2 .$$

The Husimi function gives the distribution of the evolved quantum state over the spin phase space, where each point (θ, ϕ) corresponds to a spin coherent state. Thus, by plotting $H_n(\theta, \phi)$ at different values of n, the evolution of the initially localized quantum state can be followed directly in phase space.

The Husimi evolution is then compared with the corresponding classical trajectory. In this way, the three initial states allow us to examine how quantum states initially localized in a large regular island, a smaller regular island chain, and the chaotic sea evolve under the same Floquet dynamics.

Husimi plots for evolution of state in small regular island $(0.70, 0.70, -0.16)$

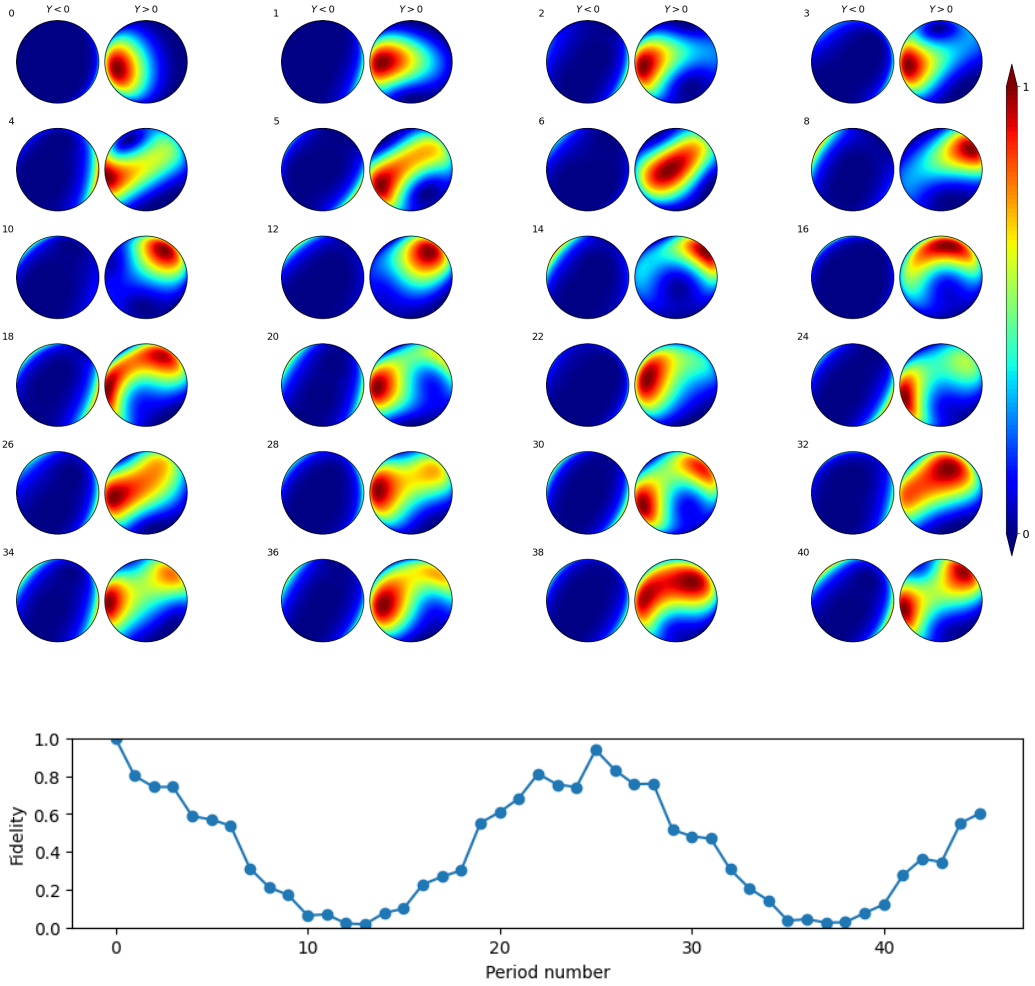


Figure 3.4: a) Husimi plot of 0 to 40 kick evolved state corresponding to intial state (0.70,0.70,-0.16). b) Fidelity at each kick of the same initial condition.

The figure shows the evolution of the spin coherent state initially located at $(X, Y, Z) = (0.70, 0.70, -0.16)$, which lies in the small regular-island region of the classical phase space. Classically, this initial condition belongs to a five-period island chain and the trajectory remains confined within the small regular island region. This confinement is associated with the invariant structures of the regular island, which prevent a classical trajectory from crossing into the surrounding chaotic sea. Such invariant structures are closely related to the invariant tori described by the KAM theory. Therefore, for this initial

condition, the chaotic sea is classically inaccessible. The quantum evolution, however, shows a different behaviour. As the state evolves, the Husimi distribution can be seen to move from the initially occupied small island towards the other small regular island in the same $Y > 0$ hemisphere through the classically forbidden chaotic region. This indicates quantum dynamical tunnelling between the two classically separated regular island, a process that has no corresponding classical trajectory.

The fidelity shown below the Husimi evolution provides further evidence for this behaviour. Here, the variation in the kicking strength is not included, so the evolution is purely under the Floquet operator with k = 2.0. The repeated rise and fall of the fidelity indicates that the quantum state periodically returns close to its initial state after leaving it. This periodic behaviour is consistent with tunnelling between the two small regular islands depicted in the husimi plots. Thus, the state isperiodically transferring between the two classically separated regions.

Husimi plots for evolution of state in Large regular island $(0.0, -0.99, -0.16)$

The figure shows the evolution of the spin coherent state initially located at $(X, Y, Z) = (0.0, -0.99, -0.16)$, which lies in the large regular-island region of the classical phase space. In contrast to the tunnelling observed for the small regular island, the Husimi distribution remains localized in the same region of phase space throughout the evolution, with only small changes in its shape and position. This behaviour is also reflected in the fidelity, which remains relatively high throughout the evolution, indicating that the evolved state retains a significant overlap with the initial state. This quantum behaviour is consistent with the corresponding classical dynamics, where the trajectory remains confined to the large regular island and forms a closed invariant curve. Thus, both the classical trajectory and the quantum state remain localized within the same regular region of phase space.

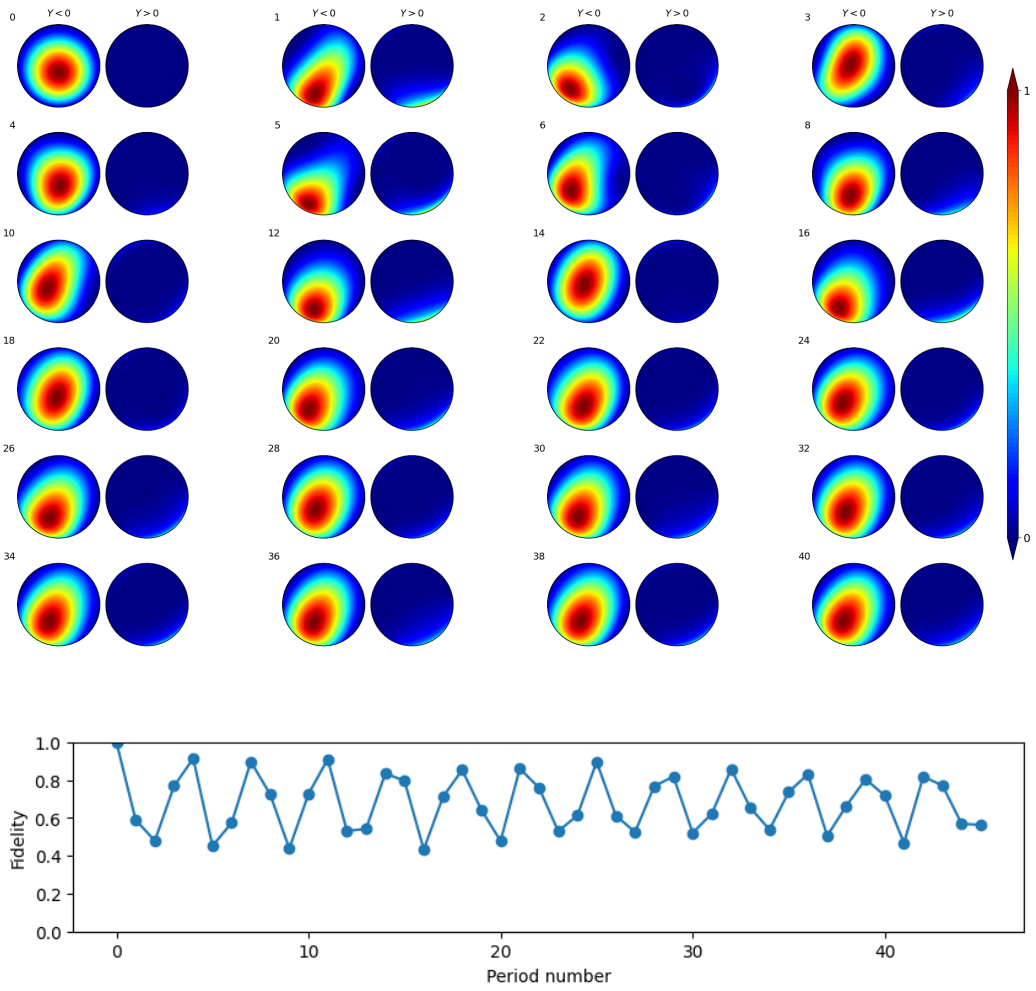


Figure 3.5: a) Husimi plot of 0 to 40 kick evolved state corresponding to intial state (0.0,-0.99,-0.16). b) Fidelity at each kick of the same initial condition.

Husimi plots for evolution of state in chaotic sea $(-0.94, 0.31, -0.16)$

The figure shows the evolution of the spin coherent state initially located in the chaotic sea at $(X, Y, Z) = (-0.94, 0.31, -0.16)$. In contrast to the states initially placed in the regular regions, the Husimi distribution does not remain confined to a particular island. Instead, it evolves through the regions of phase space which is almost similar to the classical chaotic sea more. The distribution changes significantly from one kick to the next, indicating that the quantum state is not localized around a single regular structure. This behaviour is consistent with the corresponding classical dynamics, where a

trajectory starting in the chaotic sea explores the surrounding phase-space region rather than remaining confined to a closed invariant curve. The fidelity also shows predominantly low values with irregular fluctuations, indicating that the evolved quantum state rapidly loses its overlap with the initial state.

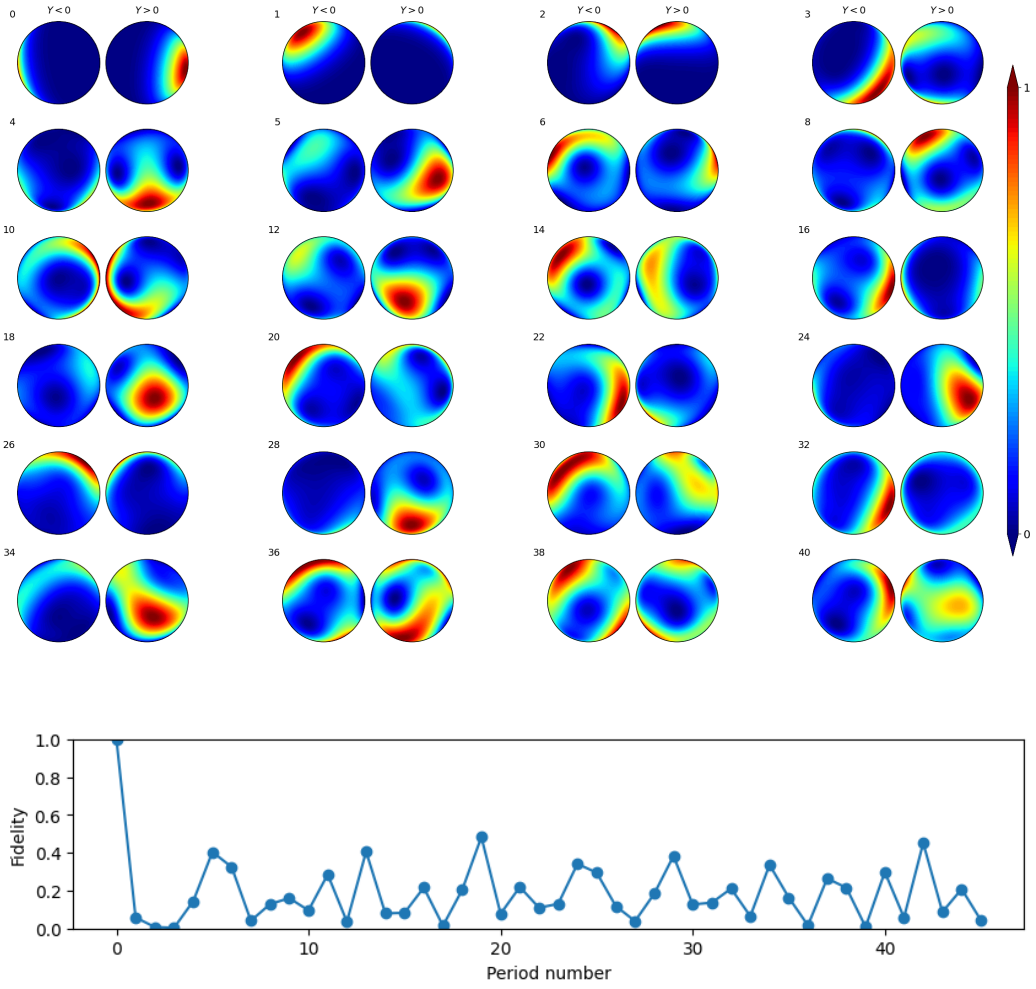


Figure 3.6: a) Husimi plot of 0 to 40 kick evolved state corresponding to intial state (-0.94,0.31,-0.16). b) Fidelity at each kick of the same initial condition.

The figure shows the evolution of the spin coherent state initially located in the chaotic sea at $(X, Y, Z) = (-0.94, 0.31, -0.16)$. In contrast to the states initially placed in the regular regions, the Husimi distribution does not remain confined to a particular island. Instead, it evolves through the regions of phase space which is almost similar to the classical chaotic sea more. The

distribution changes significantly from one kick to the next, indicating that the quantum state is not localized around a single regular structure. This behaviour is consistent with the corresponding classical dynamics, where a trajectory starting in the chaotic sea explores the surrounding phase-space region rather than remaining confined to a closed invariant curve. The fidelity also shows predominantly low values with irregular fluctuations, indicating that the evolved quantum state rapidly loses its overlap with the initial state.

40-Kick Averaged Phase Space

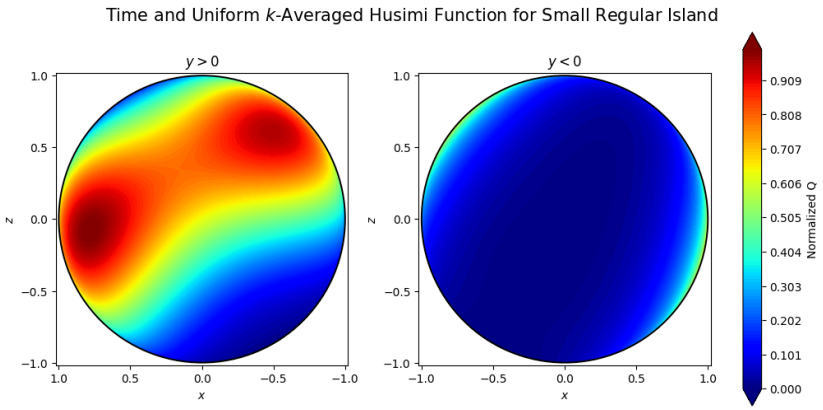
To obtain a clearer picture of the regions explored by each quantum state, we now consider the Husimi distribution averaged over the first 40 kicks. Rather than examining the Husimi function at individual time steps, the average distribution provides an overall representation of the phase-space region visited by the state during this period. This allows us to compare the quantum dynamics of the three initial states and relate their averaged distributions to the corresponding classical phase-space structures.

For an initial state $|\psi_0\rangle$, the Husimi function after the n th kick is denoted by $H_n(\theta, \phi)$. The 40-kick averaged Husimi distribution is therefore defined as

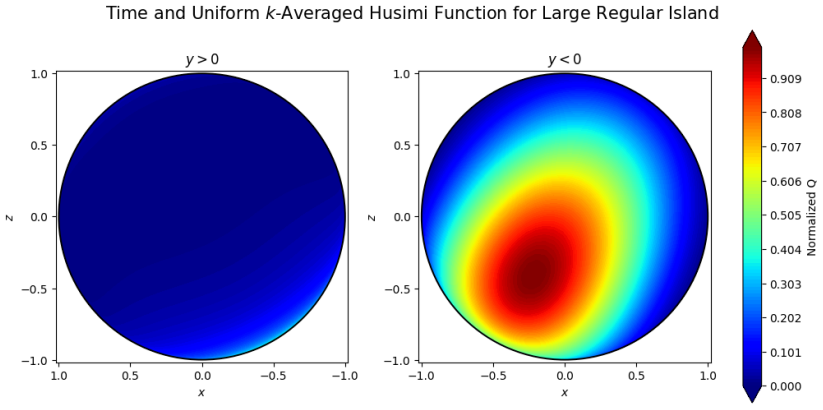
$$\overline{H}_{40}(\theta, \phi) = \frac{1}{40} \sum_{n=1}^{40} H_n(\theta, \phi).$$

The resulting distribution gives the average phase-space weight of the quantum state over the first 40 kicks. In this way, regions where the state spends more time appear more prominently, while regions visited only briefly contribute less to the averaged distribution.

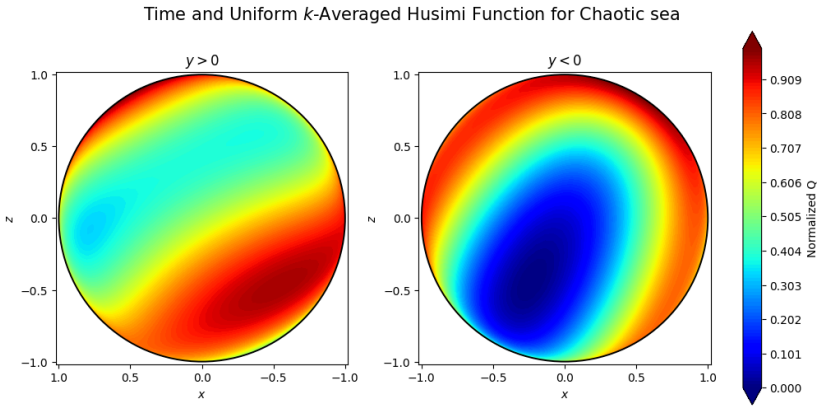
The 40-kick averaged Husimi distributions further highlight the differences between the three types of dynamics. For the state initially located in the small regular-island region, the distribution is mainly confined to the two small regular islands. This reflects the tunnelling observed in the individual Husimi plots, where the state periodically transfers between the two islands but does not spread significantly into the rest of the phase space. For the state initially located in the large regular island, the averaged distribution remains strongly localized around the same regular region, showing that the state explores only a small part of the available phase space. In contrast, the state initially placed in the chaotic sea develops a broad distribution over a large portion of the phase space. However, the large regular island remains almost unoccupied, consistent with the classical separation between the chaotic sea and the regular island. The state also has significant weight in the region containing the two smaller regular islands, indicating that this region



(a) 40 period average for the initial state (0.70,0.70,-0.16)



(b) 40 period average for the initial state (-0.94,0.31,-0.16)



(c) 40 period average for the initial state (0.00,-0.99,-0.16)

Figure 3.7: 40 period averaged Husimi plots

is accessible to the quantum state during its evolution. Thus, the averaged Husimi distributions provide a clear distinction between the confinement of regular states and the much broader phase-space exploration associated with the chaotic initial state.

Chapter 4

Discussion

4.1 Floquet Eigenstate Decomposition and Quantum Dynamics

The evolution of an arbitrary quantum state can be understood conveniently in terms of the Floquet eigenstates. The Floquet eigenstates $|e_i\rangle$ satisfy

$$U_F |e_i\rangle = e^{-i\Phi_i} |e_i\rangle ,$$

where ϕ_i is the corresponding eigenphase. Since the Floquet eigenstates form a complete basis of the Hilbert space, any initial state $|\psi_0\rangle$ can be written as a linear combination of these eigenstates:

$$|\psi_0\rangle = \sum_i c_i |e_i\rangle ,$$

where

$$c_i = \langle e_i|\psi_0\rangle.$$

Thus, the magnitude squared of the coefficient,

$$p_n = |c_i|^2 = |\langle e_i|\psi_0\rangle|^2,$$

represents the contribution of the ith Floquet eigenstate to the initial state.

To see how these contributions determine the evolution, we apply the Floquet operator to the initial state:

$$|\psi_1\rangle = U_F |\psi_0\rangle = U_F \sum_i c_i |e_i\rangle .$$

Using the eigenvalue equation for each Floquet eigenstate, we obtain

$$|\psi_1\rangle = \sum_i c_i e^{-i\Phi_i} |e_i\rangle .$$

After n kicks, this becomes

$$|\psi_n\rangle = U_F^n |\psi_0\rangle = \sum_i c_i e^{-in\Phi_i} |e_i\rangle .$$

This expression shows that the magnitude of each Floquet-state coefficient does not change during the evolution. In fact,

$$\left|c_i e^{-in\Phi_i}\right|^2 = |c_i|^2,$$

since $e^{-in\phi_i}$ has unit magnitude. Therefore, the Floquet evolution does not change the weight of each eigenstate in the initial state. Instead, each eigenstate acquires a phase that depends on its eigenphase ϕ_i.

Although the individual overlaps with the Floquet eigenstates remain constant,

$$|\langle e_i|\psi_n\rangle|^2 = |c_i|^2,$$

the relative phases between different eigenstate components change with each kick. These changing relative phases affect the interference between the different components and therefore determine the observable evolution of the state.

Table 4.1: Floquet eigenphases and overlaps of the three initial states with the corresponding Floquet eigenstates.

Floquet States	Eigenphase	p_i (large island)	p_i (small island)	p_i (chaotic sea)
$\|e_1\rangle$	-2.7835	0.81	0.00	0.03
$\|e_2\rangle$	-2.4539	0.00	0.39	0.13
$\|e_3\rangle$	-2.1985	0.00	0.53	0.07
$\|e_4\rangle$	-1.0176	0.13	0.00	0.11
$\|e_5\rangle$	0.5146	0.01	0.01	0.21
$\|e_6\rangle$	1.8257	0.03	0.04	0.26
$\|e_7\rangle$	2.8802	0.00	0.03	0.19

4.1.1 Interpretation of the Dynamics from Floquet Eigenstate Overlaps

The overlap of the three initial states with the Floquet eigenstates provides a useful way to understand the different quantum dynamics observed above. From Table 4.1, the initial state chosen in the large regular island has a dominant overlap with $|e_1\rangle$, with $p_1 = 0.81$, while the contribution from

the remaining states is relatively small. Therefore, the initial state can be approximately written as

$$|\psi_0\rangle \approx c_1 |e_1\rangle .$$

Since $|e_1\rangle$ is a Floquet eigenstate, its evolution is simply

$$U_F^n |e_1\rangle = e^{-in\Phi_1} |e_1\rangle .$$

Thus, the state evolves approximately as

$$|\psi_n\rangle \approx c_1 e^{-in\Phi_1} |e_1\rangle .$$

The factor $e^{-in\phi_1}$ is only a global phase and does not change the physical state. Therefore, the quantum state remains approximately unchanged during the evolution, which is consistent with the localized Husimi distribution and the relatively high fidelity observed for the large regular island.

The initial state chosen in the small regular-island region has a very different eigenstate decomposition. From Table 4.1, it is mainly supported by $|e_2\rangle$ and $|e_3\rangle$, with overlap probabilities $p_2 = 0.39$ and $p_3 = 0.53$, respectively. The initial state can therefore be approximated by

$$|\psi_0\rangle \approx c_2 |e_2\rangle + c_3 |e_3\rangle .$$

Its evolution after n kicks is then

$$|\psi_n\rangle \approx c_2 e^{-in\Phi_2} |e_2\rangle + c_3 e^{-in\Phi_3} |e_3\rangle .$$

Unlike the previous case, the two terms acquire different phases because their eigenphases are different. Factoring out the phase of the first term gives

$$|\psi_n\rangle \approx e^{-in\Phi_2} \left[c_2 |e_2\rangle + c_3 e^{-in(\Phi_3 - \Phi_2)} |e_3\rangle \right] .$$

The overall phase $e^{-in\Phi_2}$ has no physical effect, while the relative phase

$$\Delta\Phi = \Phi_2 - \Phi_3$$

changes with every kick. This changing relative phase produces interference between the two Floquet eigenstate components. As a result, the quantum state periodically changes its localization between the two small regular islands.

The tunnelling period can therefore be obtained directly from the difference between the two eigenphases. After N kicks, the relative phase accumulated between the two components is

$$\Delta\Phi_N = N\Delta\Phi.$$

In the idealized case of an equal superposition of the two Floquet states, the state reaches the opposite island when the relative phase changes by . Since the present state is close to, but not exactly, an equal superposition, this relation provides an approximate tunnelling period.

$$N_{\text{half}}\Delta\Phi = \pi,$$

and hence

$$N_{\text{half}} = \frac{\pi}{\Delta\Phi}.$$

A complete tunnelling cycle, in which the state returns to its original island, requires a relative phase change of 2π. Therefore, the tunnelling period is

$$N_{\text{tun}} = \frac{2\pi}{\Delta\Phi}.$$

For the present system, the relevant eigenphases are

$$\Phi_2 = -2.4539, \qquad \Phi_3 = -2.1985,$$

giving

$$\Delta\Phi = |\Phi_2 - \Phi_3| = |-2.4539 - (-2.1985)| = 0.2554.$$

Therefore, the predicted tunnelling period is

$$N_{\text{tun}} = \frac{2\pi}{0.2554} \approx 24.6 \text{ kicks},$$

and the corresponding half period is

$$N_{\text{half}} = \frac{\pi}{0.2554} \approx 12.3 \text{ kicks}.$$

This agrees well with the Husimi evolution, where the state is seen to transfer from one small regular island to the other at approximately 12 kicks. The state subsequently returns towards the original island, giving a complete tunnelling cycle of approximately 25 kicks. The fidelity also shows a corresponding periodic recurrence, supporting the interpretation of the observed dynamics as quantum tunnelling between the two classically separated regular islands.

Finally, the initial state chosen in the chaotic sea has significant overlap with all seven Floquet eigenstates. Unlike the regular-island states, its dynamics are therefore not dominated by one or two Floquet eigenstates. The evolved state contains several components with different eigenphases,

$$|\psi_n\rangle = \sum_{i=1}^{7} c_i e^{-in\Phi_i} |e_i\rangle ,$$

so that many different relative phases develop during the evolution. The resulting interference between these components changes rapidly with the number of kicks. This is consistent with the strongly varying Husimi distribution and the predominantly low and irregular fidelity observed for the chaotic initial state. Thus, the Floquet eigenstate decomposition provides a direct explanation for the different quantum dynamics associated with the large regular island, the small regular-island pair, and the chaotic sea.

Chapter 5

Conclusion

In this work, the quantum dynamics of the kicked top were studied through the correspondence between the classical phase-space structure and the quantum evolution of spin coherent states. Starting from the experimentally implemented Hamiltonian, the corresponding Floquet operator was obtained and the classical map was derived in the semiclassical limit. The resulting classical phase-space portrait showed a mixed phase space containing both regular and chaotic regions. Three initial conditions were then selected from a large regular island, a smaller regular-island chain, and the chaotic sea in order to compare their classical and quantum dynamics.

The quantum evolution of these initial states was studied using spin coherent states and the Husimi Q-function. The state initially located in the large regular island remained strongly localized in the same region throughout the evolution, consistent with the corresponding classical trajectory being confined to a closed invariant curve. In contrast, the state initially located in the small regular-island chain showed periodic transfer between the two classically separated regular islands. This behaviour has no classical counterpart and provides a clear signature of quantum dynamical tunnelling. The state initially placed in the chaotic sea showed a much broader evolution across the phase space, together with a rapid and irregular decrease in its fidelity.

The Floquet eigenstate decomposition provided further insight into these different behaviours. The initial state in the large regular island was found to be dominated by a single Floquet eigenstate, explaining its relatively stable evolution. The state in the small regular-island region was mainly composed of two Floquet eigenstates with different eigenphases. The relative phase accumulated between these two components leads to the periodic tunnelling observed in the Husimi evolution. From the difference between their eigen-

phases, $\Delta\Phi = 0.2554$, the tunnelling period was estimated as

$$N_{\text{tun}} = \frac{2\pi}{\Delta\Phi} \approx 24.6$$

kicks, corresponding to a half-period of approximately 12.3 kicks. This is in good agreement with the transfer between the two islands observed in the Husimi evolution. Finally, the chaotic initial state had significant overlap with several Floquet eigenstates, resulting in a more complicated evolution due to the accumulation of multiple relative phases.

The results obtained in this work show that we were able to successfully model the experimentally implemented Hamiltonian and reproduce the main qualitative features of the results reported by Chaudhury *et al.* [1] for pure states. The classical phase-space structure, Floquet eigenstates, and the quantum evolution of the three different initial conditions were found to be in good agreement with the corresponding results in the reference work. A small difference is observed in the detailed kick-by-kick evolution of the quantum states. This is mainly because the present work uses Schrödinger evolution of pure states, whereas the theoretical modelling in Chaudhury *et al.* uses a Lindblad master equation to include the effects of decoherence and other experimental imperfections. As a result, the quantum Husimi distributions and fidelity can differ slightly at individual kicks. However, when the Husimi distributions are averaged over the first 40 kicks, the resulting phase-space distributions show close qualitative agreement with the corresponding results of the reference work. This indicates that, despite the difference in the treatment of open-system effects, the Schrödinger evolution used here captures the main dynamical features of the experimentally implemented kicked-top system.

Overall, the results demonstrate how the classical phase-space structure is reflected in the quantum dynamics of the kicked top, while also revealing genuinely quantum effects such as dynamical tunnelling that are absent in the classical description. The combination of classical phase-space portraits, Husimi distributions, fidelity, and Floquet eigenstate analysis therefore provides a useful way of studying the quantum signatures of chaos in a periodically driven spin system.

Bibliography

[1] Souma Chaudhury, Anupam Smith, BE Anderson, Shohini Ghose, and Poul S Jessen. Quantum signatures of chaos in a kicked top. *Nature*, 461(7265):768–771, 2009.

[2] Steven H Strogatz. *Nonlinear dynamics and chaos: with applications to physics, biology, chemistry, and engineering*. CRC press, 2024.

[3] Arul Lakshminarayan. Classical and quantum chaos plus rmt and some applications, 2018. Notes for the Bangalore Summer School on Statistical Physics, ICTS-TIFR, 9–13 July 2018.

[4] Avadhut V. Purohit and Udaysinh T. Bhosale. Quantum kicked top: A paradigmatic model, 2026.

[5] Hai-Woong Lee. Theory and application of the quantum phase-space distribution functions. *Physics Reports*, 259(3):147–211, 1995.

[6] Eugen Merzbacher. *Quantum mechanics*. John Wiley & Sons, 1998.